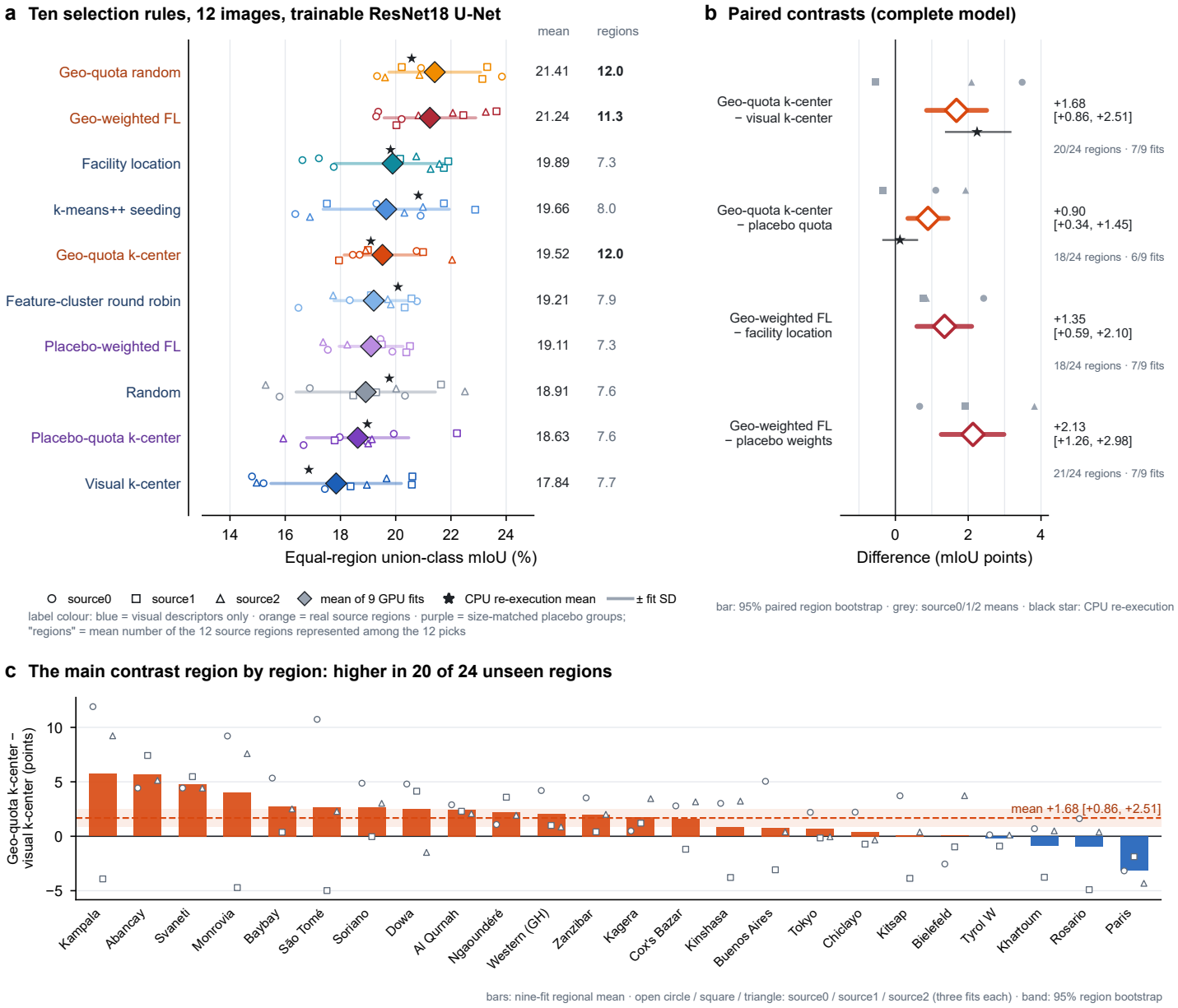

a Ten selection rules, 12 images, trainable ResNet18 U-Net
mean
regions
Geo-quota random
21.41
12.0
Geo-weighted FL
21.24
11.3
Facility location
19.89
7.3
k-means++ seeding
19.66
8.0
Geo-quota k-center
19.52
12.0
Feature-cluster round robin
19.21
7.9
Placebo-weighted FL
19.11
7.3
Random
18.91
7.6
Placebo-quota k-center
18.63
7.6
Visual k-center
17.84
7.7
14
16
18
20
22
24
Equal-region union-class mIoU (%)
b Paired contrasts (complete model)
Geo-quota k-center − visual k-center
+1.68 [+0.86, +2.51]
20/24 regions · 7/9 fits
Geo-quota k-center − placebo quota
+0.90 [+0.34, +1.45]
18/24 regions · 6/9 fits
Geo-weighted FL − facility location
+1.35 [+0.59, +2.10]
18/24 regions · 7/9 fits
Geo-weighted FL − placebo weights
+2.13 [+1.26, +2.98]
21/24 regions · 7/9 fits
0
2
4
Difference (mIoU points)
source0
source1
source2
mean of 9 GPU fits
CPU re-execution mean
± fit SD
label colour: blue = visual descriptors only · orange = real source regions · purple = size-matched placebo groups;
"regions" = mean number of the 12 source regions represented among the 12 picks
bar: 95% paired region bootstrap · grey: source0/1/2 means · black star: CPU re-execution
c The main contrast region by region: higher in 20 of 24 unseen regions
Geo-quota k-center − visual k-center (points)
10
5
0
−5
mean +1.68 [+0.86, +2.51]
Kampala
Abancay
Svaneti
Monrovia
Baybay
São Tomé
Soriano
Dowa
Al Qurnah
Ngaoundéré
Western (GH)
Zanzibar
Kagera
Cox's Bazar
Kinshasa
Buenos Aires
Tokyo
Chiclayo
Kitsap
Bielefeld
Tyrol W
Khartoum
Rosario
Paris
bars: nine-fit regional mean · open circle / square / triangle: source0 / source1 / source2 (three fits each) · band: 95% region bootstrap